



Name: Gargi Sukhatankar
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EXPERIMENT NO. 5

Aim: To implement SQL lite browser to view data on android.

SQLite Browser: SQLite Browser (also known as DB Browser for SQLite) is a free, open-source, visual tool designed to simplify working with SQLite database files. It provides an intuitive graphical interface that allows users to create, browse, edit, and run SQL queries on SQLite databases without writing code. Ideal for developers and testers, it makes inspecting and managing local app data straightforward, especially during mobile app development or debugging.

Analysis:

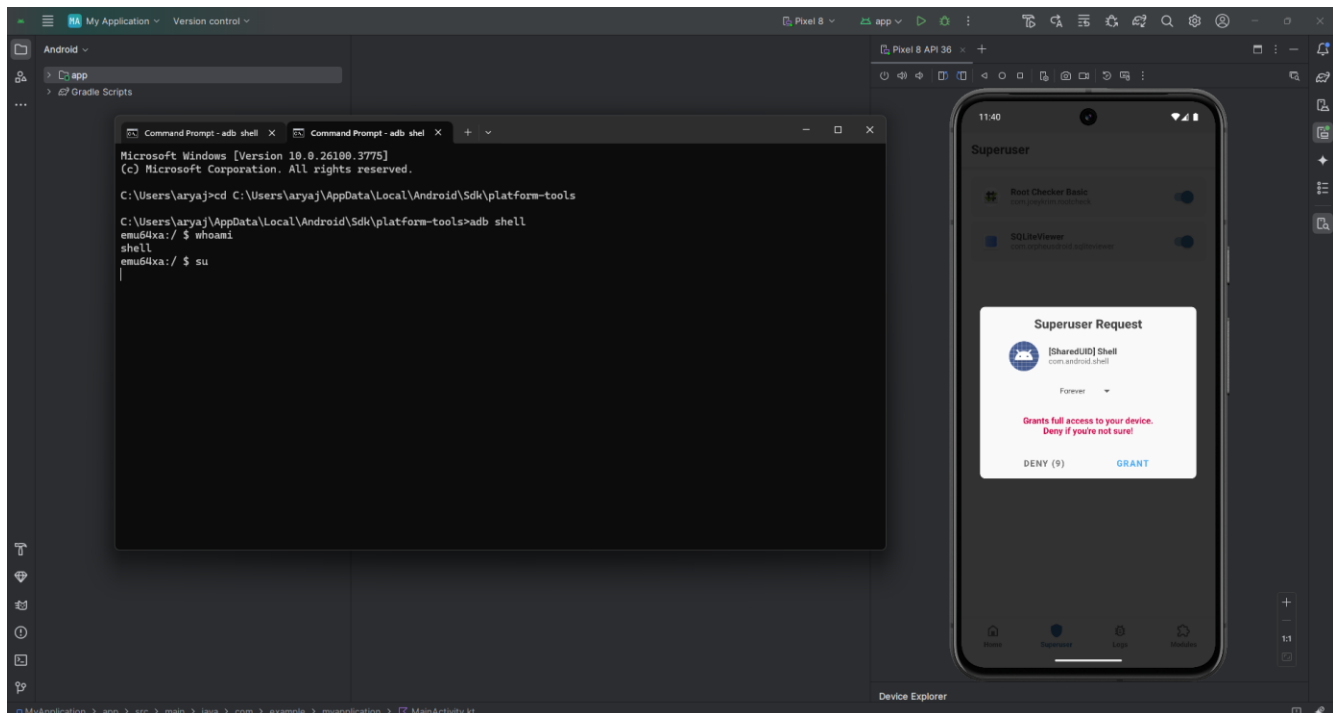


Figure 1: Granting Root Permissions



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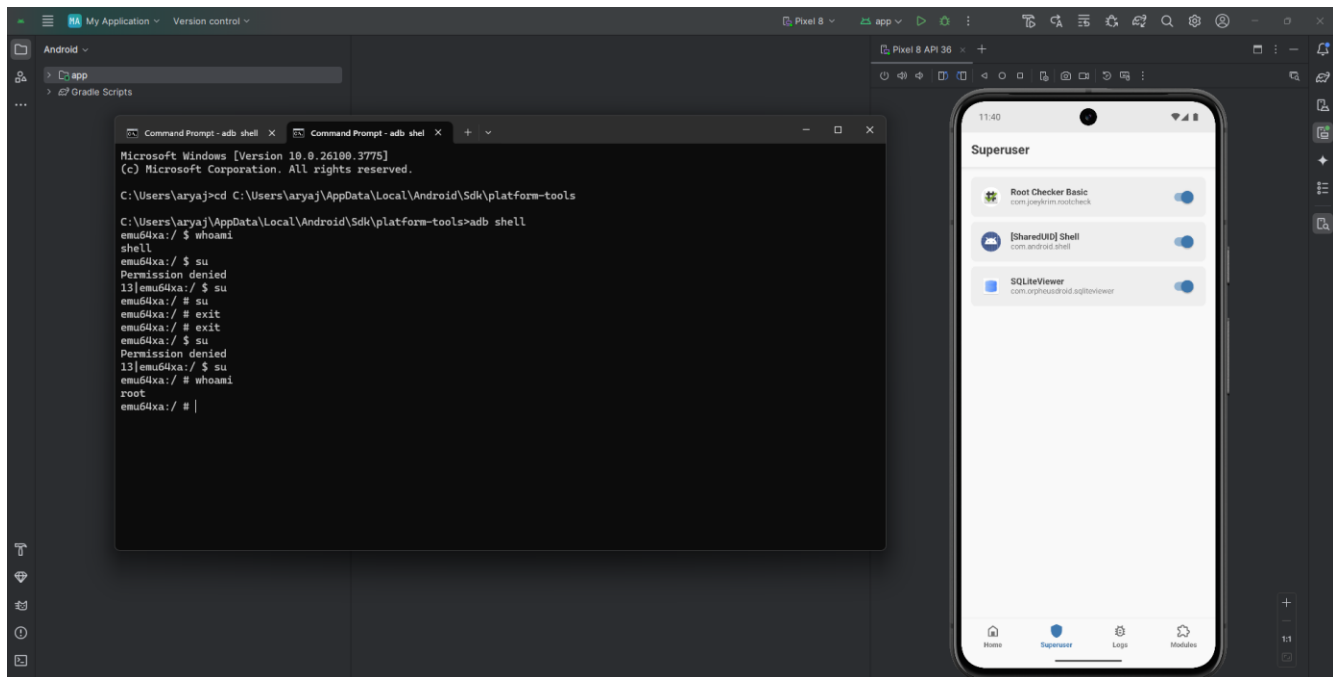


Figure 2: Successfully Granted Root Permissions



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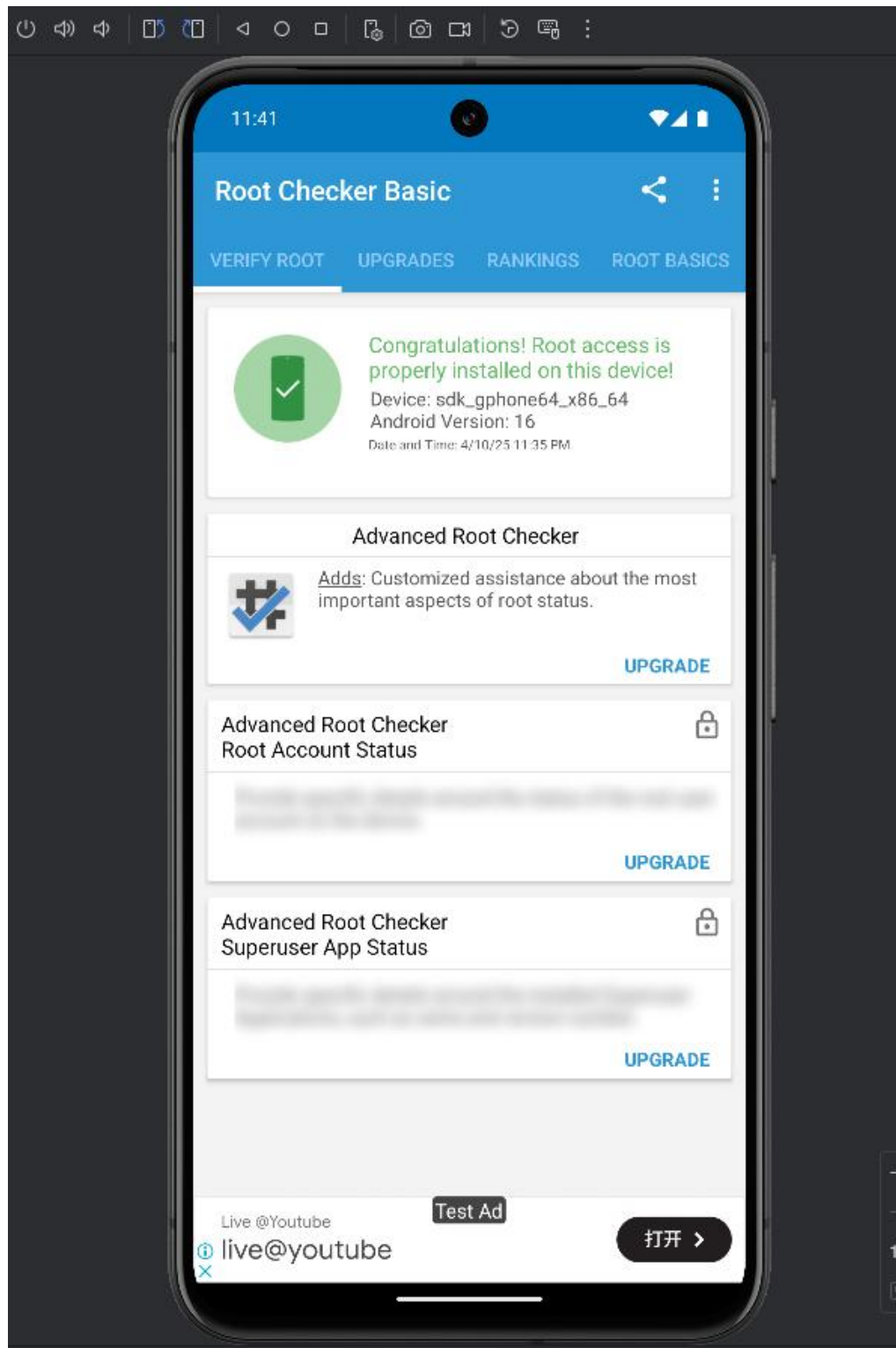


Figure 3: Root Checker



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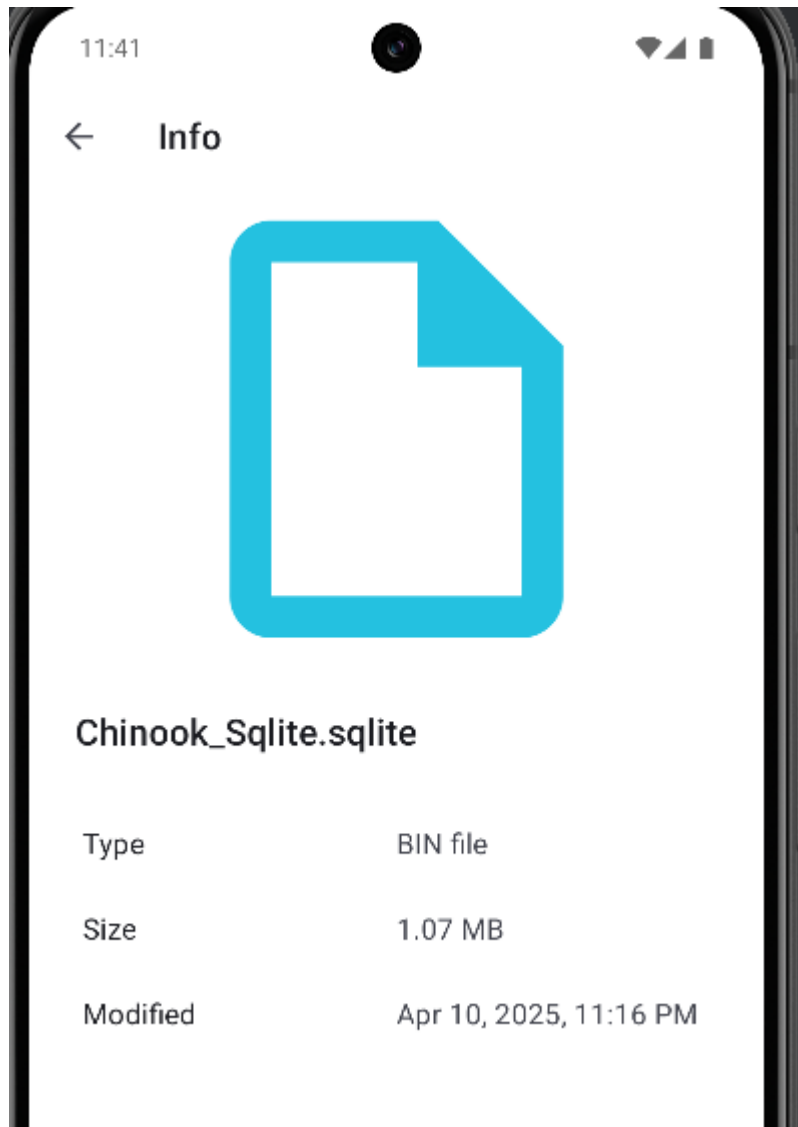


Figure 3: SQLite File created



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D8 Browser for SQLite - C:\Users\arya\Downloads\Chinook_Sqlite.sqlite

File Edit View Tools Help

New Database Open Database Write Changes Revert Changes Undo Open Project Save Project Attach Database Close Database

Database Structure Browse Data Edit Pgmas Execute SQL

Create Table Create Index Modify Table Delete Table Print Refresh

Name	Type	Schema
Tables (11)		
Album		CREATE TABLE [Album] ([AlbumId] INTEGER NOT NULL, [Title] NVARCHAR(160) NOT NULL, [ArtistId] INTEGER NOT NULL, CONSTRAINT [PK_Album] PRIMARY KEY ([AlbumId]), FOREIGN KEY ([ArtistId]) REFER
Artist		CREATE TABLE [Artist] ([ArtistId] INTEGER NOT NULL, [Name] NVARCHAR(120), CONSTRAINT [PK_Artist] PRIMARY KEY ([ArtistId]))
Customer		CREATE TABLE [Customer] ([CustomerId] INTEGER NOT NULL, [FirstName] NVARCHAR(40) NOT NULL, [LastName] NVARCHAR(20) NOT NULL, [Company] NVARCHAR(80), [Address] NVARCHAR(70), [City] NVA
Employee		CREATE TABLE [Employee] ([EmployeeId] INTEGER NOT NULL, [LastName] NVARCHAR(20) NOT NULL, [FirstName] NVARCHAR(20) NOT NULL, [Title] NVARCHAR(30), [ReportsTo] INTEGER, [BirthDate] DATETI
Genre		CREATE TABLE [Genre] ([GenreId] INTEGER NOT NULL, [Name] NVARCHAR(120), CONSTRAINT [PK_Genre] PRIMARY KEY ([GenreId]))
Invoice		CREATE TABLE [Invoice] ([InvoiceId] INTEGER NOT NULL, [CustomerId] INTEGER NOT NULL, [InvoiceDate] DATETIME NOT NULL, [BillingAddress] NVARCHAR(70), [BillingCity] NVARCHAR(40), [BillingState] NVA
InvoiceLine		CREATE TABLE [InvoiceLine] ([InvoiceLineId] INTEGER NOT NULL, [InvoiceId] INTEGER NOT NULL, [TrackId] INTEGER NOT NULL, [UnitPrice] NUMERIC(10,2) NOT NULL, [Quantity] INTEGER NOT NULL, CONSTRA
MediaType		CREATE TABLE [MediaType] ([MediaTypeId] INTEGER NOT NULL, [Name] NVARCHAR(120), CONSTRAINT [PK_MediaType] PRIMARY KEY ([MediaTypeId]))
Playlist		CREATE TABLE [Playlist] ([PlaylistId] INTEGER NOT NULL, [Name] NVARCHAR(120), CONSTRAINT [PK_Playlist] PRIMARY KEY ([PlaylistId]))
PlaylistTrack		CREATE TABLE [PlaylistTrack] ([PlaylistId] INTEGER NOT NULL, [TrackId] INTEGER NOT NULL, CONSTRAINT [PK_PlaylistTrack] PRIMARY KEY ([PlaylistId], [TrackId]), FOREIGN KEY ([PlaylistId]) REFERENCES [Pl
Track		CREATE TABLE [Track] ([TrackId] INTEGER NOT NULL, [Name] NVARCHAR(200) NOT NULL, [AlbumId] INTEGER, [MediaTypeId] INTEGER NOT NULL, [GenreId] INTEGER, [Composer] NVARCHAR(220), [Mills
Indices (21)		
IFK_AlbumArtistId		CREATE INDEX [IFK_AlbumArtistId] ON [Album] ([ArtistId])
IFK_CustomerSupportRepId		CREATE INDEX [IFK_CustomerSupportRepId] ON [Customer] ([SupportRepId])
IFK_EmployeeReportsTo		CREATE INDEX [IFK_EmployeeReportsTo] ON [Employee] ([ReportsTo])
IFK_InvoiceCustomerId		CREATE INDEX [IFK_InvoiceCustomerId] ON [Invoice] ([CustomerId])
IFK_InvoiceLineInvoiceId		CREATE INDEX [IFK_InvoiceLineInvoiceId] ON [InvoiceLine] ([InvoiceId])
IFK_InvoiceLineTrackId		CREATE INDEX [IFK_InvoiceLineTrackId] ON [InvoiceLine] ([TrackId])
IFK_PlaylistTrackTrackId		CREATE INDEX [IFK_PlaylistTrackTrackId] ON [PlaylistTrack] ([TrackId])
IFK_TrackAlbumId		CREATE INDEX [IFK_TrackAlbumId] ON [Track] ([AlbumId])
IFK_TrackGenreId		CREATE INDEX [IFK_TrackGenreId] ON [Track] ([GenreId])
IFK_TrackMediaTypeId		CREATE INDEX [IFK_TrackMediaTypeId] ON [Track] ([MediaTypeId])
IFK_Album		CREATE UNIQUE INDEX [IFK_Album] ON [Album]([AlbumId])
IFK_Artist		CREATE UNIQUE INDEX [IFK_Artist] ON [Artist]([ArtistId])
IFK_Customer		CREATE UNIQUE INDEX [IFK_Customer] ON [Customer]([CustomerId])
IFK_Employee		CREATE UNIQUE INDEX [IFK_Employee] ON [Employee]([EmployeeId])
IFK_Genre		CREATE UNIQUE INDEX [IFK_Genre] ON [Genre]([GenreId])
IFK_Invoice		CREATE UNIQUE INDEX [IFK_Invoice] ON [Invoice]([InvoiceId])
IFK_InvoiceLine		CREATE UNIQUE INDEX [IFK_InvoiceLine] ON [InvoiceLine]([InvoiceLineId])
IFK_MediaType		CREATE UNIQUE INDEX [IFK_MediaType] ON [MediaType]([MediaTypeId])
IFK_Playlist		CREATE UNIQUE INDEX [IFK_Playlist] ON [Playlist]([PlaylistId])
IFK_PlaylistTrack		CREATE UNIQUE INDEX [IFK_PlaylistTrack] ON [PlaylistTrack]([PlaylistId], [TrackId])
IFK_Track		CREATE UNIQUE INDEX [IFK_Track] ON [Track]([TrackId])
Views (0)		
Triggers (0)		

Name	Type	Schema
Tables (11)		
Album		CREATE TABLE [Album] ([AlbumId] INTEGER NOT NULL, [Title] NVARCHAR(160) NOT NULL, [ArtistId] INTEGER NOT NULL, CONSTRAINT [PK_Album] PRIMARY KEY ([AlbumId]), FOREIGN KEY ([ArtistId]) REF
AlbumId	INTEGER	"AlbumId" INTEGER NOT NULL
Title	NVARCHAR(1...	"Title" NVARCHAR(160) NOT NULL
ArtistId	INTEGER	"ArtistId" INTEGER NOT NULL
Artist		CREATE TABLE [Artist] ([ArtistId] INTEGER NOT NULL, [Name] NVARCHAR(120), CONSTRAINT [PK_Artist] PRIMARY KEY ([ArtistId]))
ArtistId	INTEGER	"ArtistId" INTEGER NOT NULL
Name	NVARCHAR(1...	"Name" NVARCHAR(120)
Customer		CREATE TABLE [Customer] ([CustomerId] INTEGER NOT NULL, [FirstName] NVARCHAR(40) NOT NULL, [LastName] NVARCHAR(20) NOT NULL, [Company] NVARCHAR(80), [Address] NVARCHAR(70), [City] N
CustomerId	INTEGER	"CustomerId" INTEGER NOT NULL
FirstName	NVARCHAR(40)	"FirstName" NVARCHAR(40) NOT NULL
LastName	NVARCHAR(20)	"LastName" NVARCHAR(20) NOT NULL
Company	NVARCHAR(80)	"Company" NVARCHAR(80)
Address	NVARCHAR(70)	"Address" NVARCHAR(70)
City	NVARCHAR(40)	"City" NVARCHAR(40)
State	NVARCHAR(40)	"State" NVARCHAR(40)
Country	NVARCHAR(40)	"Country" NVARCHAR(40)
PostalCode	NVARCHAR(10)	"PostalCode" NVARCHAR(10)
Phone	NVARCHAR(24)	"Phone" NVARCHAR(24)
Fax	NVARCHAR(24)	"Fax" NVARCHAR(24)
Email	NVARCHAR(60)	"Email" NVARCHAR(60) NOT NULL
SupportRepId	INTEGER	"SupportRepId" INTEGER
Employee		CREATE TABLE [Employee] ([EmployeeId] INTEGER NOT NULL, [LastName] NVARCHAR(20) NOT NULL, [FirstName] NVARCHAR(20) NOT NULL, [Title] NVARCHAR(30), [ReportsTo] INTEGER, [BirthDate] DATE
Genre		CREATE TABLE [Genre] ([GenreId] INTEGER NOT NULL, [Name] NVARCHAR(120), CONSTRAINT [PK_Genre] PRIMARY KEY ([GenreId]))
Invoice		CREATE TABLE [Invoice] ([InvoiceId] INTEGER NOT NULL, [CustomerId] INTEGER NOT NULL, [InvoiceDate] DATETIME NOT NULL, [BillingAddress] NVARCHAR(70), [BillingCity] NVARCHAR(40), [BillingState] N
InvoiceId	INTEGER	"InvoiceId" INTEGER NOT NULL
CustomerId	INTEGER	"CustomerId" INTEGER NOT NULL
InvoiceDate	DATETIME	"InvoiceDate" DATETIME NOT NULL
BillingAddress	NVARCHAR(70)	"BillingAddress" NVARCHAR(70)
BillingCity	NVARCHAR(40)	"BillingCity" NVARCHAR(40)
BillingState	NVARCHAR(40)	"BillingState" NVARCHAR(40)
BillingCountry	NVARCHAR(40)	"BillingCountry" NVARCHAR(40)
BillingPostalCode	NVARCHAR(10)	"BillingPostalCode" NVARCHAR(10)
Total	NUMERIC(10, 2)	"Total" NUMERIC(10, 2) NOT NULL
InvoiceLine		CREATE TABLE [InvoiceLine] ([InvoiceLineId] INTEGER NOT NULL, [InvoiceId] INTEGER NOT NULL, [TrackId] INTEGER NOT NULL, [UnitPrice] NUMERIC(10,2) NOT NULL, [Quantity] INTEGER NOT NULL, CONST
MediaType		CREATE TABLE [MediaType] ([MediaTypeId] INTEGER NOT NULL, [Name] NVARCHAR(120), CONSTRAINT [PK_MediaType] PRIMARY KEY ([MediaTypeId]))
Playlist		CREATE TABLE [Playlist] ([PlaylistId] INTEGER NOT NULL, [Name] NVARCHAR(120), CONSTRAINT [PK_Playlist] PRIMARY KEY ([PlaylistId]))
PlaylistTrack		CREATE TABLE [PlaylistTrack] ([PlaylistId] INTEGER NOT NULL, [TrackId] INTEGER NOT NULL, CONSTRAINT [PK_PlaylistTrack] PRIMARY KEY ([PlaylistId], [TrackId]), FOREIGN KEY ([PlaylistId]) REFERENCES
Track		CREATE TABLE [Track] ([TrackId] INTEGER NOT NULL, [Name] NVARCHAR(200) NOT NULL, [AlbumId] INTEGER, [MediaTypeId] INTEGER NOT NULL, [GenreId] INTEGER, [Composer] NVARCHAR(220), [Mills
Indices (21)		
IFK_AlbumArtistId		CREATE INDEX [IFK_AlbumArtistId] ON [Album] ([ArtistId])

Figure 5: Reading SQLite File using SQLite Browser



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Conclusion:

Successfully Rooted Android Emulator and used SQLite browser to view database file for Android Apk.